

DEPARTMENT OF ELECTRICAL ENGINEERING

Electronic Devices & Circuits Laboratory (EDC LAB)

EXPERIMENT 1: V–I Characteristics of PN Junction & Zener Diodes, and Study of Clipper & Clamper Circuits

Prepared By: Suraj.N (EE25BTECH11056)
Sai Krishna Bakki (EE25BTECH11049)
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1. Aim

1. To plot the forward- and reverse-bias V – I characteristics of a PN junction diode and determine its cut-in voltage (V_γ), static forward resistance (R_{DC}), dynamic forward resistance (r_d), and reverse saturation current (I_s).
2. To plot the forward- and reverse-bias V – I characteristics of a Zener diode (1N4733A) and determine its Zener breakdown voltage (V_Z) and Zener resistance (R_Z).
3. To design, simulate, and test positive, negative, and biased (two-level) diode clipper circuits and observe their clipping levels.
4. To design, simulate, and test positive and negative diode clamper circuits and measure the resulting DC voltage shift.

2. Apparatus and Software Tools Required

Table 1: Equipment and Components Specifications

S.No.	Component / Equipment	Specification	Quantity
1	PN Junction Diode	1N4007 / 1N4148	1
2	Zener Diode	1N4733A ($V_Z = 5.1$ V, 1 W)	1
3	Resistors	1 k Ω , 10 k Ω , 100 k Ω	As required
5	DC Voltage Sources	0–30 V Variable, +1 V, –1 V Reference	As required
6	Function Generator / AC Source	1 kHz Sine wave, 4 V _{pp}	1
7	Simulation Software	LTspice	–
8	Digital Storage Oscilloscope (DSO)	Hardware measurement	1

3. Part A: PN Junction Diode Characteristics

3.1 Theoretical Background

A PN junction diode exhibits non-linear current-voltage characteristics described by the Shockley diode equation:

$$I = I_s \left(e^{\frac{V}{\eta V_T}} - 1 \right) \quad (1)$$

where I_s is the reverse saturation current, $V_T = \frac{kT}{q}$ is the thermal voltage (≈ 26 mV at room temperature), and η is the ideality factor.

- **Forward Bias:** Below cut-in voltage ($V_\gamma \approx 0.6\text{--}0.7$ V for Silicon), current is negligible. Above V_γ , current increases exponentially.
- **Reverse Bias:** Barrier height increases, allowing only minute reverse saturation current I_s to flow until reverse breakdown occurs.

3.2 Circuit Diagrams and Simulation Waveforms

3.2.1 PN Junction Diode – Forward Bias

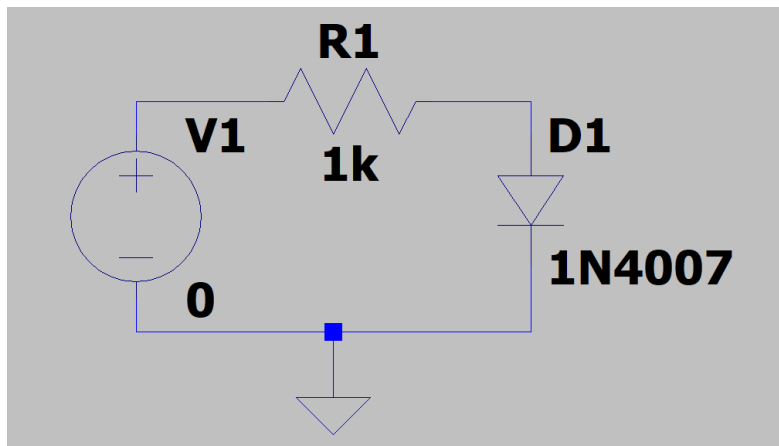
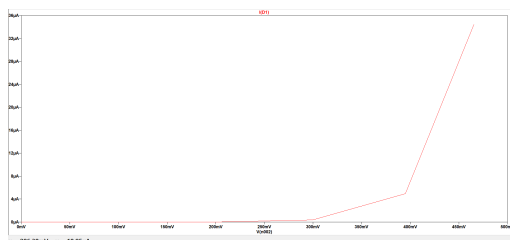
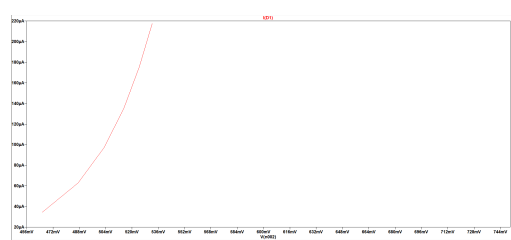


Figure 1: PN Junction Diode Forward Bias Circuit Schematic.



(a) Low Voltage Sweep (0 V to 0.5 V)



(b) Knee Region Sweep (0.5 V to 0.75 V)

Figure 2: PN Diode Forward V - I Characteristics Simulation Plots.

3.2.2 PN Junction Diode – Reverse Bias

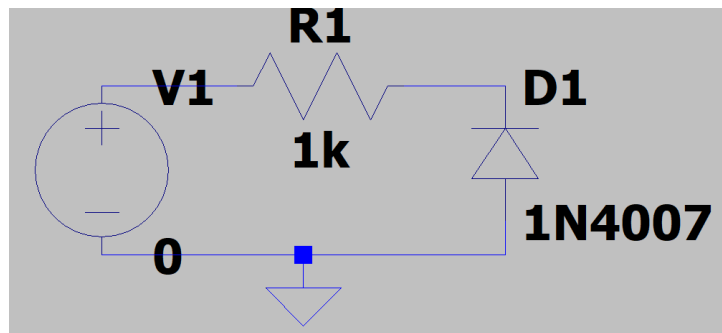


Figure 3: PN Junction Diode Reverse Bias Circuit Schematic.

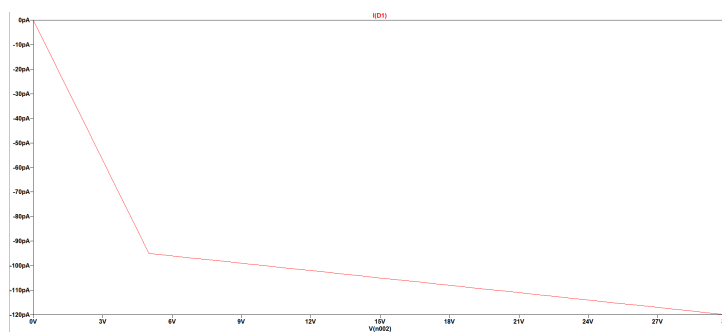


Figure 4: PN Diode Reverse Characteristic (0 V to 30 V).

3.3 Experimental Observations (PN Diode)

Table 2: PN Junction Diode (1N4007) Forward-Bias Readings

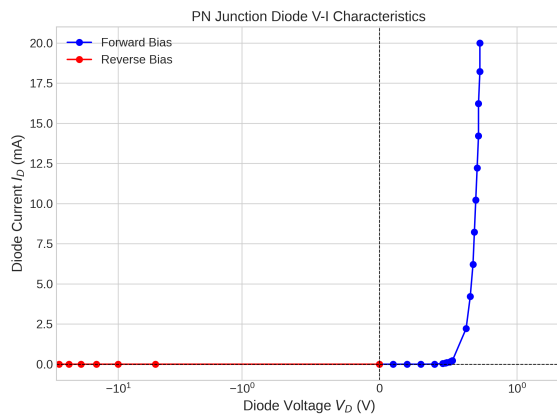
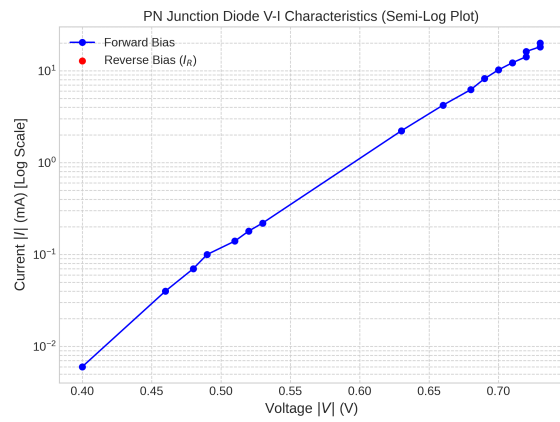
S.No.	V_{sup} (V)	V_D (V)	I_D (mA)
1	0.00	0.00	0.000
2	0.10	0.10	0.000
3	0.20	0.20	0.000
4	0.30	0.30	0.000
5	0.40	0.40	0.006
6	0.50	0.46	0.040
7	0.55	0.48	0.070
8	0.60	0.49	0.100
9	0.65	0.51	0.140
10	0.70	0.52	0.180
11	0.75	0.53	0.220
12	2.88	0.63	2.220
13	4.93	0.66	4.220
14	6.97	0.68	6.220
15	9.01	0.69	8.220
16	11.04	0.70	10.220
17	13.08	0.71	12.220
18	15.11	0.72	14.220
19	17.14	0.72	16.220
20	19.17	0.73	18.220
21	20.97	0.73	20.000

Table 3: PN Junction Diode (1N4007) Reverse-Bias Readings

S.No.	V_R (V)	I_R (μ A)
1	0	0
2	-5	0
3	-10	0
4	-15	0
5	-20	0
6	-25	0
7	-30	0

3.4 Experimental V-I Characteristics & Plots

The experimental readings were plotted using Matplotlib in both linear scale (I vs V) and semi-log scale ($\log |I|$ vs V).

(a) Linear Scale Plot (I vs V)(b) Semi-Log Plot ($\log |I|$ vs V)Figure 5: Experimental V - I Characteristic Curves for PN Junction Diode (1N4007).

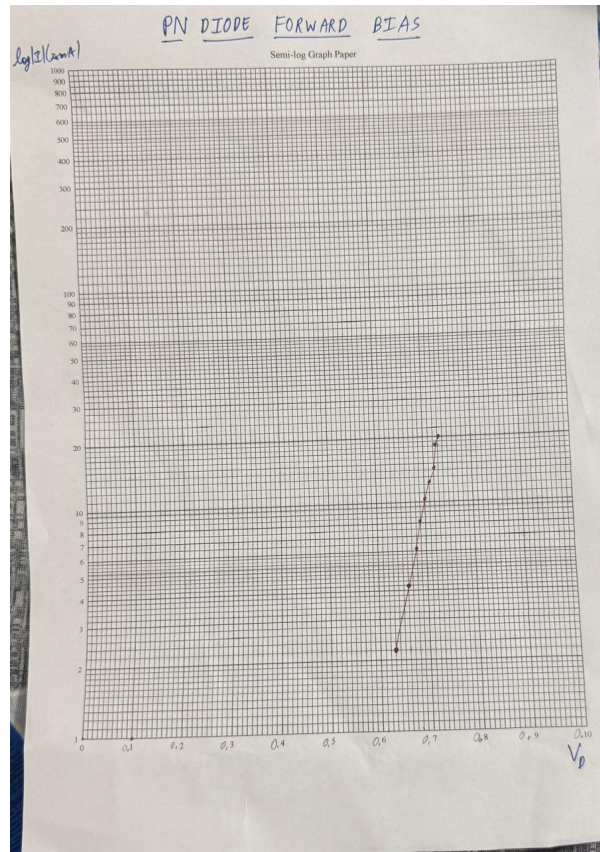


Figure 6: Graph-Paper Plot of the Forward-Bias V - I Characteristics of PN Junction Diode.

3.5 Calculated Parameters (PN Diode)

- **Cut-in Voltage (V_γ):** ≈ 0.60 V (exponential current rise begins above 0.6 V).
- **Static Forward Resistance ($R_{DC} = V_f/I_f$):**
 - At $I_f = 2.22$ mA, $V_f = 0.63$ V: $R_{DC} = \frac{0.63}{2.22 \text{ mA}} = 283.78 \text{ } \Omega$.
 - At $I_f = 4.22$ mA, $V_f = 0.66$ V: $R_{DC} = \frac{0.66}{4.22 \text{ mA}} = 156.40 \text{ } \Omega$.
 - At $I_f = 10.22$ mA, $V_f = 0.70$ V: $R_{DC} = \frac{0.70}{10.22 \text{ mA}} = 68.49 \text{ } \Omega$.
 - At $I_f = 20.0$ mA, $V_f = 0.73$ V: $R_{DC} = \frac{0.73}{20.0 \text{ mA}} = 36.50 \text{ } \Omega$.
- **Dynamic Forward Resistance ($r_d = \Delta V_f / \Delta I_f$):**
 - In knee region (2.22 mA to 4.22 mA): $r_d = \frac{0.66 - 0.63}{4.22 \text{ mA} - 2.22 \text{ mA}} = \frac{0.03 \text{ V}}{2 \text{ mA}} = 15.00 \text{ } \Omega$.
 - In high-conduction region (10.22 mA to 20.0 mA): $r_d = \frac{0.73 - 0.70}{20.0 \text{ mA} - 10.22 \text{ mA}} = \frac{0.03 \text{ V}}{9.78 \text{ mA}} = 3.07 \text{ } \Omega$.
- **Reverse Saturation Current:** The measured reverse current remained $0 \text{ } \mu\text{A}$ from 0 V to -30 V; hence I_s was below the DMM resolution and no reverse breakdown was observed in the tested range.

4. Part B: Zener Diode Characteristics

4.1 Theoretical Background

A Zener diode is designed to operate safely in reverse breakdown (V_Z). Once in breakdown, the terminal voltage remains virtually constant over a wide range of reverse currents. In forward bias, it behaves as a normal PN junction diode.

4.2 Circuit Diagrams and Simulation Waveforms

4.2.1 Zener Diode – Forward Bias

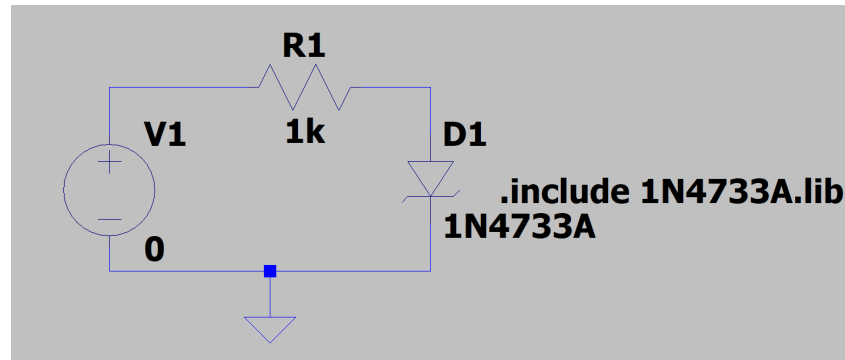
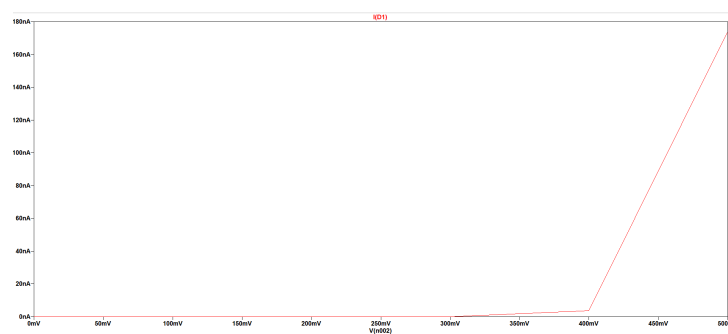
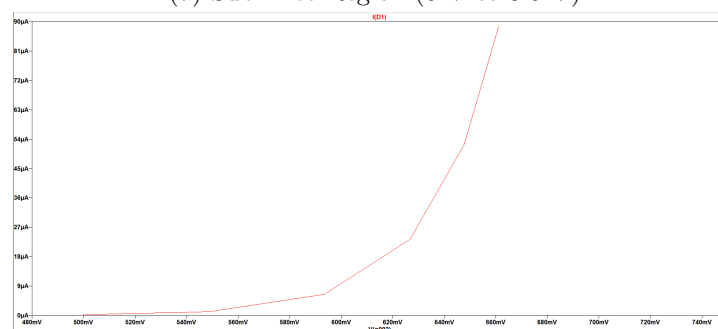


Figure 7: Zener Diode Forward Bias Circuit Schematic.



(a) Sub-knee Region (0 V to 0.5 V)



(b) Conduction Region (0.5 V to 0.75 V)

Figure 8: Zener Diode Forward Bias Simulation Plots.

4.2.2 Zener Diode – Reverse Bias and Breakdown

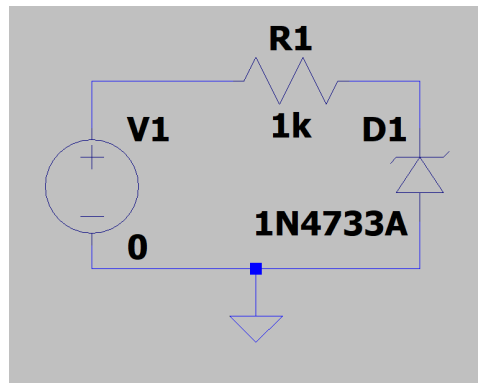


Figure 9: Zener Diode Reverse Bias Circuit Schematic.

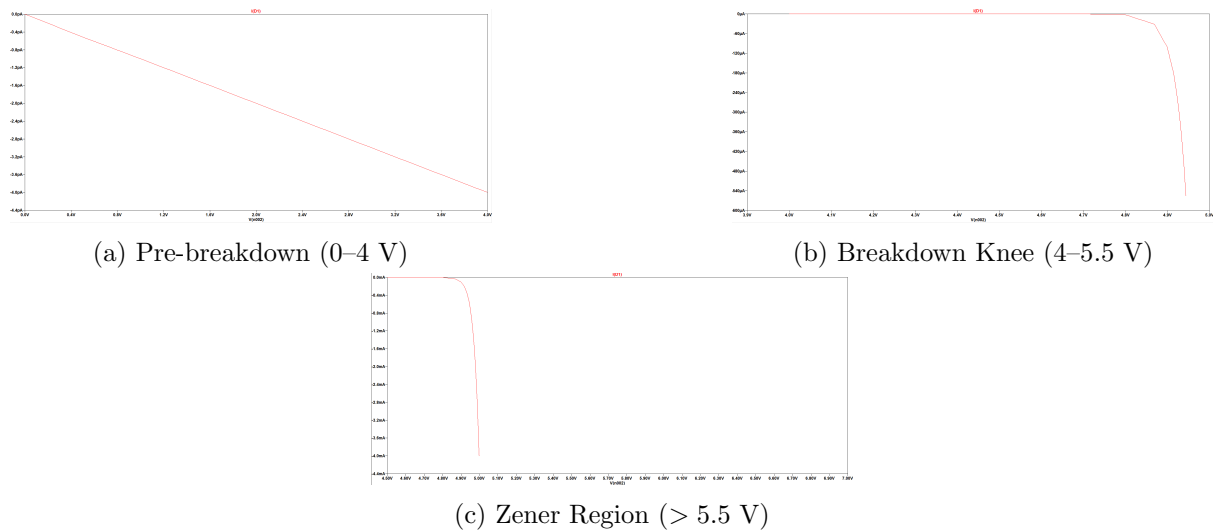


Figure 10: Zener Diode Reverse Breakdown Simulation Response.

4.3 Experimental Observations (Zener Diode)

Table 4: Zener Diode (1N4733A) Forward-Bias Readings

S.No.	V_{sup} (V)	V_Z (V)	I_Z (mA)
1	0.00	0.00	0.000
2	0.10	0.10	0.000
3	0.20	0.20	0.000
4	0.30	0.30	0.000
5	0.40	0.40	0.000
6	0.50	0.50	0.000
7	0.55	0.55	0.002
8	0.60	0.59	0.008
9	0.65	0.62	0.025
10	0.70	0.65	0.050
11	0.75	0.66	0.090

Table 5: Zener Diode (1N4733A) Reverse-Bias Readings

S.No.	V_{sup} (V)	V_Z (V)	I_Z (mA)
1	-0.00	-0.00	0.000
2	-0.50	-0.50	0.000
3	-1.00	-1.00	0.000
4	-1.50	-1.50	0.000
5	-2.00	-2.00	0.000
6	-2.50	-2.50	-0.001
7	-3.00	-3.00	-0.005
8	-3.50	-3.48	-0.020
9	-4.00	-3.93	-0.080
10	-4.10	-4.00	-0.100
11	-4.20	-4.08	-0.120
12	-4.30	-4.15	-0.150
13	-4.40	-4.22	-0.180
14	-4.50	-4.29	-0.220
15	-4.60	-4.35	-0.260
16	-4.70	-4.40	-0.300
17	-4.80	-4.45	-0.350
18	-4.90	-4.50	-0.410
19	-5.00	-4.54	-0.460
20	-5.10	-4.58	-0.520
21	-5.20	-4.62	-0.590
22	-5.30	-4.65	-0.650
23	-5.40	-4.68	-0.720
24	-5.50	-4.70	-0.800
25	-7.82	-5.00	-2.800
26	-9.93	-5.09	-4.800
27	-11.99	-5.14	-6.800
28	-14.04	-5.17	-8.800
29	-16.08	-5.19	-10.800
30	-18.12	-5.21	-12.800
31	-20.15	-5.22	-14.800
32	-22.18	-5.23	-16.800
33	-24.21	-5.24	-18.800
34	-25.43	-5.25	-20.000

4.4 Experimental V-I Characteristics & Plots

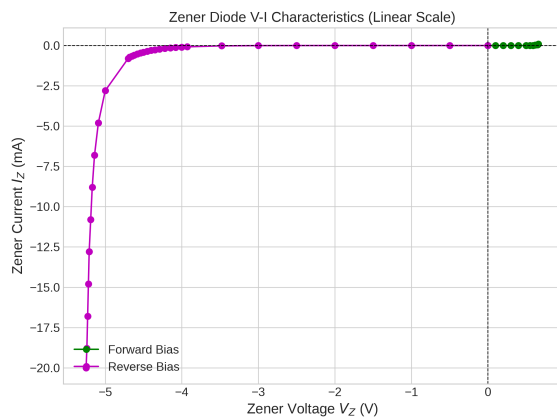
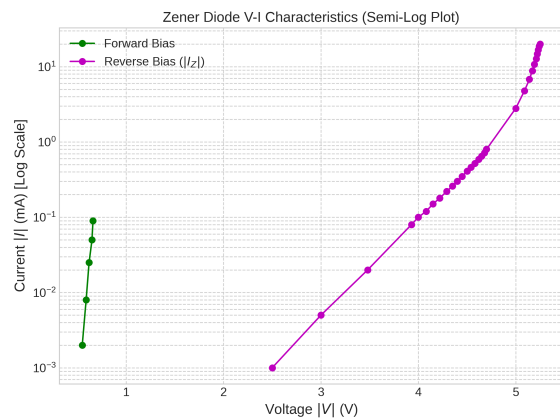
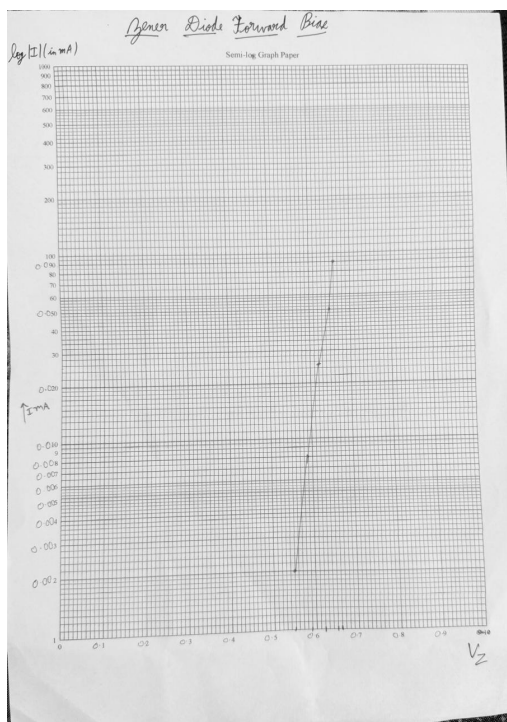
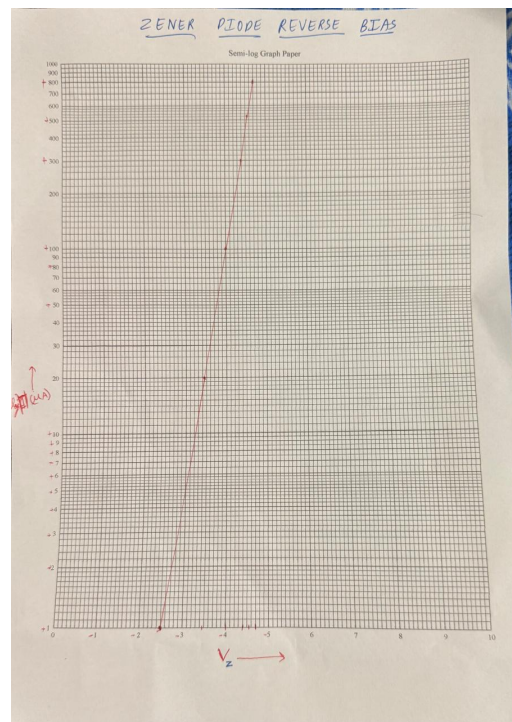
(a) Linear Scale Plot (I vs V)(b) Semi-Log Plot ($\log |I|$ vs V)Figure 11: Experimental V - I Characteristic Curves for Zener Diode (1N4733A).(a) Graph-Paper Plot of the Forward-Bias V - I Characteristics.(b) Graph-Paper Plot of the Reverse-Bias V - I Characteristics.

Figure 12: Graph-Paper Plots for Zener Diode (1N4733A).

4.5 Calculated Parameters (Zener Diode)

- **Zener Breakdown Voltage (V_Z):** Breakdown starts at $V_Z \approx 5.00$ V and stabilizes up to 5.25 V at 20 mA.
- **Zener Dynamic Resistance ($R_Z = \Delta V_Z / \Delta I_Z$):**

$$R_Z = \frac{5.25 \text{ V} - 5.00 \text{ V}}{20.0 \text{ mA} - 2.8 \text{ mA}} = \frac{0.25 \text{ V}}{17.2 \text{ mA}} = 14.53 \Omega \quad (2)$$

5. Part C: Diode Clipper Circuits

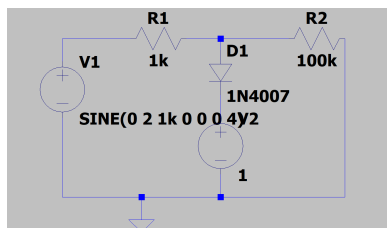
5.1 Theoretical Background

Clippers remove portions of an applied signal above or below chosen reference levels without distorting the remainder. A load resistor of $R_L = 100\text{ k}\Omega$ was connected across the output to ensure smooth DSO readings and eliminate loading effects.

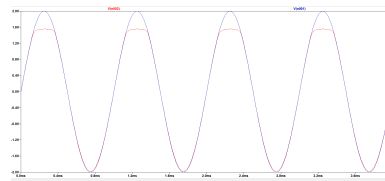
5.2 Circuit Diagrams, Simulations & Hardware DSO Waveforms

5.2.1 Positive Clipper (Clipping above +1 V reference)

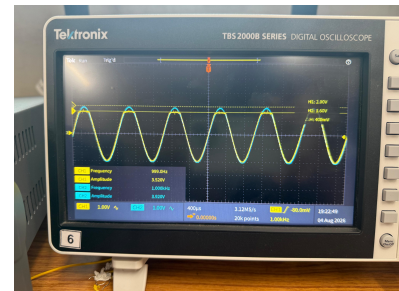
In positive clipping, the diode conducts when $V_{in} > +1\text{ V} + V_\gamma$, holding output to $+1.60\text{ V}$.



(a) Circuit Schematic



(b) LTspice Simulation

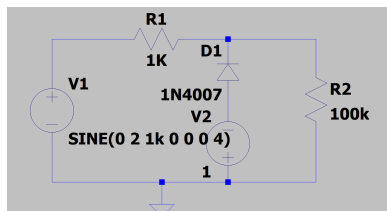


(c) DSO Trace

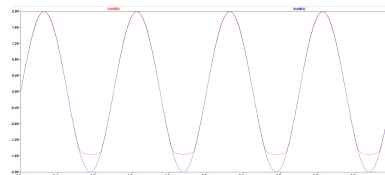
Figure 13: Positive Clipper Circuit, Simulation, and Measured DSO Waveform (Clipped at $+1.60\text{ V}$).

5.2.2 Negative Clipper (Clipping below -1 V reference)

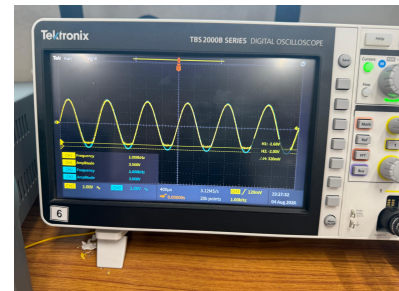
In negative clipping, the diode conducts when $V_{in} < -1\text{ V} - V_\gamma$, holding output to -1.68 V .



(a) Circuit Schematic



(b) LTspice Simulation

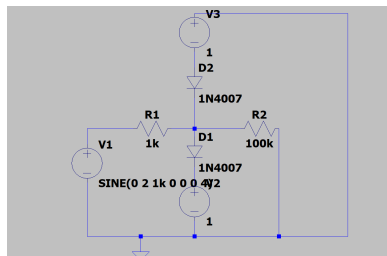


(c) DSO Trace

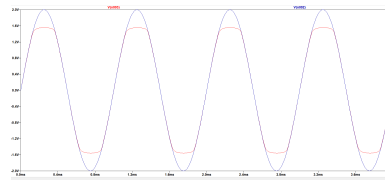
Figure 14: Negative Clipper Circuit, Simulation, and Measured DSO Waveform (Clipped at -1.68 V).

5.2.3 Biased Clipper (Two-Level Clipper)

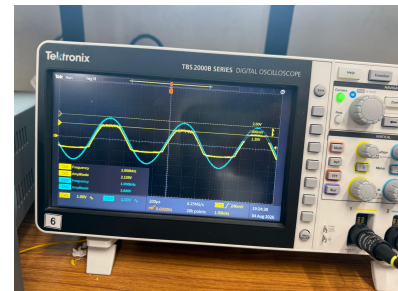
Both positive and negative peaks are clipped independently. The upper limit clipped at $+1.20\text{ V}$ and lower limit clipped at -0.92 V .



(a) Circuit Schematic



(b) LTspice Simulation



(c) DSO Trace

Figure 15: Biased Clipper Circuit, Simulation, and Measured DSO Waveform (Clipped at $+1.20\text{ V}$ & -0.92 V).

5.3 Clipper Experimental Observation Summary

Table 6: Diode Clipper Circuits Observation Table ($R_L = 100\text{ k}\Omega$)

Circuit	Input V_{pp} (V)	Measured Clip Level	Inference
Positive Clipper	4.0	$+1.60\text{ V}$	Upper peak clipped cleanly above $V_{ref} + V_\gamma$.
Negative Clipper	4.0	-1.68 V	Lower peak clipped cleanly below $-V_{ref} - V_\gamma$.
Biased Clipper	4.0	$+1.20\text{ V}$ & -0.92 V	Dual-level clipping limits peak-to-peak swing.

6. Part D: Diode Clamper Circuits

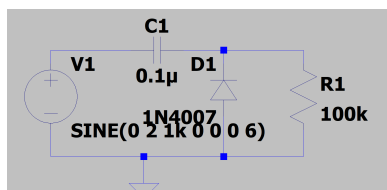
6.1 Theoretical Background

A clamper shifts the DC level of an AC waveform without changing its peak-to-peak value significantly. The diode and capacitor charge during one half-cycle and add a DC offset during the other half-cycle, producing positive or negative clamping depending on the diode orientation.

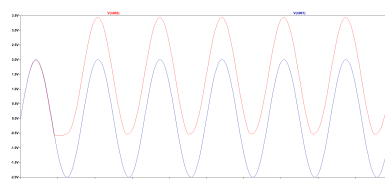
6.2 Circuit Diagrams, Simulations & Hardware DSO Waveforms

6.2.1 Positive Clamper

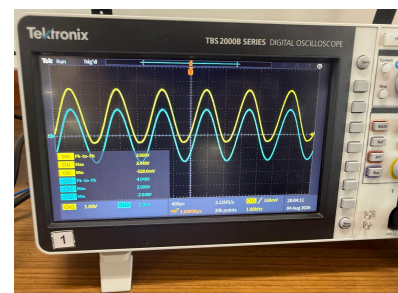
The positive clamper shifts the waveform upward such that the negative peak is clamped near $-V_\gamma = -0.52\text{ V}$.



(a) Circuit Schematic



(b) LTspice Simulation



(c) DSO Trace

Figure 16: Positive Clamper Circuit, Simulation, and Measured DSO Waveform (Min level at -0.52 V).

6.2.2 Negative Clamper

The negative clamper shifts the waveform downward such that the positive peak is clamped near $+V_\gamma = +0.56$ V.

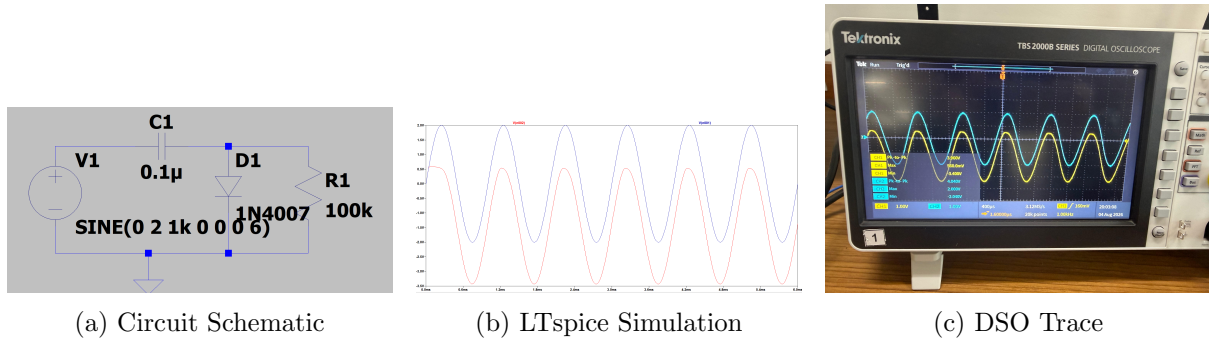


Figure 17: Negative Clamper Circuit, Simulation, and Measured DSO Waveform (Max level at $+0.56$ V).

6.3 Clamper Experimental Observation Summary

Table 7: Diode Clamper Circuits Observation Table ($R_L = 100$ k Ω)

Circuit	Input V_{pp} (V)	Theoretical Shift	Measured Clamped Level
Positive Clamper	4.0	Shift up by $+V_p - V_\gamma \approx +1.30$ V	Min level at -0.52 V
Negative Clamper	4.0	Shift down by $-V_p + V_\gamma \approx -1.30$ V	Max level at $+0.56$ V

Inference : in **Positive Clamper** : Upward DC shift; min level clamped at $-V_\gamma$.

In **Negative Clamper** : Downward DC shift; max level clamped at $+V_\gamma$.

7. Discussion & Overall Inferences

- PN Diode Non-linearity & Conduction:** Forward bias readings demonstrate exponential growth after overcoming cut-in barrier voltage $V_\gamma \approx 0.60$ V. Semi-log plots confirm Shockley's relation with linear slope in forward conduction. Static resistance decreases from $R_{DC} = 283.78$ Ω at 2.22 mA to $R_{DC} = 36.50$ Ω at 20.0 mA, while dynamic resistance settles at $r_d = 3.07$ Ω .
- Zener Breakdown Regulation:** In reverse bias, breakdown occurs sharply at $V_Z \approx 5.00$ – 5.25 V. The low dynamic Zener resistance $R_Z = 14.53$ Ω confirms excellent voltage regulation stability under varying reverse currents.
- Clipper Behavior:** Incorporating a 100 k Ω load resistor prevented signal attenuation. Clipping thresholds closely match $V_{ref} \pm V_\gamma$ ($+1.60$ V for positive, -1.68 V for negative, and $+1.20$ V / -0.92 V for biased clipper).
- Clamper DC Restoration:** Positive and negative clampers successfully restore the DC level, shifting the minimum peak to -0.52 V and maximum peak to $+0.56$ V respectively without altering the peak-to-peak amplitude.